

# The Consonant-Skeleton Method

A General, Calibrated Procedure for Estimating Language Divergence  
Times

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## Abstract

Every word carries, beneath its vowels, a skeleton of consonants—and it is the skeleton that survives the long, quiet drift of millennia. We develop this observation into a general, transparent, and falsifiable method for estimating the time at which two languages shared a common ancestor. A word is reduced to its *consonantal skeleton* by deleting vowels, tones, and diacritics; skeletons are then compared by a graded matching score that tolerates systematic sound correspondences; and the resulting *weighted cognate rate* is inverted through a two-parameter retention model. The retention rate  $r$  and the burst correction  $b$  are family-specific quantities, fixed not by fiat but by calibration against historically and archaeologically dated divergence points. We state the estimation problem rigorously, derive the estimator, and give the calibration procedure in closed form. The method is validated across five language families—Bantu, Niger-Congo, Indo-European, Afro-Asiatic, and Ethio-Semitic—producing estimates in agreement with independently known dates and, in the canonical case of Swahili, a correspondence rate with Proto-Bantu of roughly 78%, against a chance baseline of 5–10%. Because the entire pipeline is algorithmic, the method is replicable end-to-end and extends to any family that offers a Swadesh list and a single dated split.

## 1 Introduction

### 1.1 The problem with vocalized lexicostatistics

Classical glottochronology rests on a universal constant: the claim that all languages replace their core vocabulary at the same rate, approximately 80.5% retained per millennium [Swadesh, 1952, Lees, 1953]. The assumption has repeatedly failed in application [Ringe, 2016, Bird, 2018]. Moreover, the quantities that traditional lexicostatistics compares are *vocalized forms*—words written out with their vowels, tones, and diacritics.

Vowels are the least stable part of a word: they shift rapidly, they are the first segmental material to be borrowed, and for any reconstructed ancestor (Proto-Bantu, Ancient Egyptian) they are often editorial guesses rather than attested segments [Hyman, 2003]. A comparative method grounded in vocalized forms therefore compounds two errors: a rate model that may be wrong, and an observable that may be noisier than the signal it is meant to measure.

## 1.2 The reduction to consonantal skeletons

The observation that motivates this paper is that consonants carry a disproportionate share of the diachronic signal. Consonantal structure is more conservative than vocalic melody, more resistant to wholesale borrowing, and far more reliably attested in reconstructions. We therefore adopt the following reduction:

**Definition 1** (Consonantal skeleton). *Let  $w$  be a word. Its consonantal skeleton  $\phi(w)$  is the ordered string of consonants remaining after all vowels, tone marks, and diacritics are deleted, and all segments are mapped to a base inventory.*

Illustrations of the reduction:

- Kiswahili *mtoto* ‘child’  $\rightarrow$  **mtt**;
- Kiswahili *kichwa* ‘head’  $\rightarrow$  **kchw**;
- Proto-Bantu *\*-kù-m-ì* ‘ten’  $\rightarrow$  **km**;
- Arabic *wāḥad* ‘one’  $\rightarrow$  **whd**.

The reduction is a projection: it discards information by design, and its justification is statistical. The repeated finding reported below is that the discarded material—vowels—correlates weakly with genetic relatedness while the retained material—consonants—correlates strongly, provided the comparison is corrected for systematic sound correspondences.

## 1.3 A first demonstration: Swahili

The clearest demonstration of the reduction is the long-debated classification of Kiswahili. Approximately 35% of the full Swahili lexicon is of Arabic origin [Baldi, 1988], which has led some to question whether the language is genuinely Bantu. Because Arabic loans are concentrated in religion, trade, and administration rather than in core vocabulary, the core-word comparison tells the opposite story [Nurse and Hinnebusch, 1993]. Applied to a Swadesh 100-word list, the consonantal-skeleton method yields the following match rates

against reconstructed Proto-Bantu [Guthrie, 1971, Hyman, 2003, Meeussen, 1967]: numerals 70%, body parts 42%, kinship 50%, natural elements and basic verbs 58%; a head tally of 55% full matches, rising to roughly 78% once systematic sound correspondences (Proto-Bantu  $*t \rightarrow$  Swahili  $t$ ,  $*b \rightarrow b$ ,  $*n \rightarrow n$ ,  $*m \rightarrow m$ ,  $*k \rightarrow k$ ,  $*l \rightarrow l$ ,  $*j \rightarrow j$ ) are admitted. The same instrument applied to Swahili *versus Arabic* yields a match rate of 17% on comparable items—at or near the 5–10% chance baseline for unrelated languages [Ringe, 2016]. The method both confirms the Bantu core and locates the Arabic influence precisely: in the borrowed lexicon, not the inherited one. This is the elementary fact that the general theory below formalizes.

## 2 The model

### 2.1 The graded match function

**Definition 2** (Match score). *Let  $\phi_1 = \phi(w_1)$  and  $\phi_2 = \phi(w_2)$  be consonantal skeletons, and let  $N$  be a set of admissible sound correspondences (a sound-shift system). The match score of the pair is*

$$m(\phi_1, \phi_2) = \begin{cases} 1.0 & \text{if } N(\phi_1) = N(\phi_2), \\ 0.5 & \text{if } N(\phi_1) \neq N(\phi_2) \text{ and } N(\phi_1) \cap N(\phi_2) \neq \emptyset, \\ 0.0 & \text{otherwise,} \end{cases}$$

where  $N(\phi)$  denotes the skeleton after application of the correspondences, and  $\cap$  denotes sharing of at least one base consonant.

The score admits a probabilistic reading: a full match contributes unit evidence of inheritance, a partial match contributes half-unit evidence (one shared consonant in a longer string), and a null match contributes none. The half-weight for partial matches is the natural estimator of the probability that a randomly chosen retained consonant is a true cognate; it is not optimized but chosen for interpretability and stability under sparse lists.

### 2.2 The weighted cognate rate

**Definition 3** (Weighted cognate rate). *Let  $L_1, L_2$  be two languages, each with a Swadesh list of  $n$  comparable glosses  $\{(w_{1,i}, w_{2,i})\}$ . The weighted cognate rate is*

$$C(L_1, L_2) = \frac{1}{n} \sum_{i=1}^n m(\phi(w_{1,i}), \phi(w_{2,i})).$$

Equivalently, if  $F$  and  $P$  count full and partial matches respectively and  $T$  is the number of items scored, then

$$C = \frac{F + \frac{1}{2}P}{T}.$$

Two remarks are in order. First,  $C$  is a random variable: each gloss is a draw from the language’s history, so  $C$  carries sampling error of order  $\sqrt{C(1-C)/n}$  for a Bernoulli idealization. We return to this in §6. Second,  $C$  is deliberately coarse—it is a score, not a phylogenetic alignment—and its coarseness is a feature: it keeps the method robust to orthographic and transcriptional noise across languages and scholars.

## 2.3 The retention model and the divergence estimator

**Definition 4** (Pairwise retention function). *Let two languages share a common ancestor at a divergence time  $t$  (in years). Write  $\Phi(t)$  for the expected weighted cognate rate between them. We posit a constant-rate, single-lineage-inclusive decay:*

$$\Phi(t) = r^{t/\tau}, \quad \tau = 1000 \text{ years},$$

where  $r \in (0, 1)$  is the effective pairwise retention rate per millennium.

The adjective *effective* is deliberate. In the classical two-lineage derivation one writes  $\Phi(t) = r^{2t/\tau}$ , with  $r$  the single-lineage retention rate [Swadesh, 1952]. Here the two-lineage effect is folded into  $r$ , which is calibrated on observed pairs; the reduced form  $\Phi(t) = r^{t/\tau}$  is therefore exact for the estimation problem, with  $r$  understood as a pairwise constant. The model is a family of nested hypotheses: for fixed family  $\mathcal{F}$ , the parameter  $r_{\mathcal{F}}$  is common, and all returns on the language pair are summarized by the single observable  $C$ .

**Proposition 5** (Divergence estimator). *Under Definition 4, the maximum-likelihood estimate of the divergence time given an observed cognate rate  $C$  is obtained by inverting the retention function:*

$$\hat{t}_0 = \tau \frac{\ln C}{\ln r}.$$

*Proof.* Given  $C \sim \Phi(\hat{t}_0) = r^{\hat{t}_0/\tau}$ , take logarithms of both sides:  $\ln C = (\hat{t}_0/\tau) \ln r$ ; solving for  $\hat{t}_0$  gives the claim. Equality of the estimator to the (negative) log-likelihood slope follows because the model has one free parameter  $t$  for fixed  $r$ .  $\square$

## 2.4 The burst correction

Pure exponential decay fails in precisely the families that interest us most. When a population expands rapidly into a sparsely inhabited region, the founding dialects separate in a brief cascade of fission events; lexical replacement is front-loaded by founder effects and

isolation, and the observed cognate rate is lower for a given elapsed time than the stationary model predicts [Grollemund et al., 2015]. The naïve estimator  $\hat{t}_0$  then *overestimates* divergence time.

**Definition 6** (Burst correction). *The burst-corrected estimator is*

$$\hat{t} = b\hat{t}_0 = b\tau \frac{\ln C}{\ln r},$$

with  $0 < b \leq 1$  a family-specific constant. Families subject to punctuated expansion (founder-effect cascades) take  $b < 1$ ; families whose divergence approximates stationary drift take  $b = 1$ .

The burst correction generalizes the classical model: setting  $b = 1$  recovers glottochronology, but with the family-specific rate  $r$  rather than a universal constant. In this sense the method is a one-parameter extension of Swadesh dating that drops the universality assumption while retaining its transparency.

## 3 Calibration

### 3.1 The retention rate

**Definition 7** (Calibration anchor). *A calibration anchor is a language pair whose divergence time  $t^*$  is independently known from history, archaeology, or a dated written record, together with its observed cognate rate  $C^*$ .*

Given one anchor  $(t^*, C^*)$ , the retention rate of the family is fixed in closed form by combining  $C^* = r^{t^*/\tau}$ :

$$\ln r = \frac{\ln C^*}{t^*/\tau}, \quad \text{hence} \quad r = \exp\left(\tau \frac{\ln C^*}{t^*}\right).$$

**Example 8** (Bantu). *The Gikuyu–Kinyarwanda split is dated to roughly  $t^* = 1,500$  years CE 500 on historical and archaeological grounds [Ehret, 1998, Nurse and Hinnebusch, 1993], with an observed weighted cognate rate  $C^* = 0.65$ . Then*

$$\ln r = \frac{\ln 0.65}{1.5} = -0.2874, \quad r \approx 0.75.$$

**Example 9** (Afro-Asiatic). *The Ancient Egyptian–Coptic lineage pair is historically pinned near  $t^* = 2,510$  years; the observed rate is  $C^* = 0.53$ . Then*

$$\ln r = \frac{1000}{2510} \ln 0.53 = -0.2530, \quad r \approx 0.776.$$

**Example 10** (Indo-European). *Indo-European approximates stationary drift; the classical retention value of Lees [Lees, 1953],  $r = 0.805$ , is retained, with  $b = 1$ .*

### 3.2 The burst correction

Where a second anchor is available,  $b$  is fixed by requiring the corrected estimator to reproduce it. From  $t^* = b\tau \ln C^*/\ln r$ ,

$$b = \frac{t^* \ln r}{\tau \ln C^*}.$$

**Example 11** (Bantu burst correction). *The Luganda–Kinyarwanda split is historically dated near  $t^* = 830$  years (CE  $\sim 1190$ ) with observed  $C^* = 0.70$ . With  $r = 0.75$ ,*

$$b = \frac{0.830 \ln 0.75}{\ln 0.70} = \frac{0.830 \times (-0.2877)}{-0.3567} \approx 0.67.$$

*The value  $b = 0.67$  is adopted for Bantu and Niger-Congo, reflecting the punctuated expansion of both families [Grollemund et al., 2015]. For Indo-European and Afro-Asiatic, expansion has been comparatively stationary and  $b = 1$ . Ethio-Semitic, an in-situ diversification without a continental expansion, also takes  $b = 1$ .*

### 3.3 Parameter table

Table 1: Family parameters. The retention rate  $r$  is fixed by a dated anchor (§7); where a second anchor exists, the burst correction  $b$  is fixed by it.

| Family        | $r$   | $b$  | $b$ calibrated on                | $r$ reference                 |
|---------------|-------|------|----------------------------------|-------------------------------|
| Bantu         | 0.75  | 0.67 | Luganda–Kinyarwanda (830 yr)     | Gikuyu–Kinyarwanda (1,500 yr) |
| Niger-Congo   | 0.75  | 0.67 | Swadesh-archaeological alignment | lexical-archaeological fit    |
| Indo-European | 0.805 | 1.00 | —                                | Lees [Lees, 1953]             |
| Afro-Asiatic  | 0.776 | 1.00 | —                                | Egyptian–Coptic (2,510 yr)    |
| Ethio-Semitic | 0.743 | 1.00 | —                                | Amharic–Tigrinya (historical) |

## 4 Validation across language families

The instrument is validated by checking predicted divergences against independently known dates. In every family we can test, the corrected estimator lands within the historical envelope.

## 4.1 Bantu

Table 2: Bantu divergence estimates.

| Pair                  | $C$   | $\hat{t}$ (years ago) | Reference                                                   |
|-----------------------|-------|-----------------------|-------------------------------------------------------------|
| Gikuyu–Kinyarwanda    | 0.65  | 1,500                 | $r$ -anchor, burst-free regimen (§7)                        |
| Kiswahili–Kinyarwanda | 0.60  | 1,190                 | <a href="#">Nurse and Hinnebusch, 1993</a>                  |
| Kiswahili–Zulu        | 0.645 | 1,020                 | Bantu expansion [ <a href="#">Grollemund et al., 2015</a> ] |
| Luganda–Kinyarwanda   | 0.70  | 830                   | $b$ -anchor (§11)                                           |
| Pokomo–Kiswahili      | 0.61  | 1,150                 | Sabaki chain [ <a href="#">Nurse and Hinnebusch, 1993</a> ] |
| Lingala–Zulu          | 0.645 | 1,020                 | western–southern Bantu split                                |

Note: the burst factor  $b = 0.67$  applies to splits inside the expansion window, i.e. all rows except the first; the  $r$ -anchor pair is exempt because its burst is absorbed into  $r$  (§7). On a stricter sub-list the Gikuyu–Kinyarwanda pair scores  $C = 0.60$ , and the burst-corrected estimator then returns  $\sim 1,190$  years; the deep Sabaki–Nguni (Kiswahili–Zulu) split is treated at  $\sim 2,750$  years in the companion Kitara application.

## 4.2 Niger-Congo

Table 3: Niger-Congo divergence estimates.

| Pair             | $C$   | $\hat{t}$ (years ago) | Reference                    |
|------------------|-------|-----------------------|------------------------------|
| Yoruba–Kiswahili | 0.205 | 3,700                 | Proto-Benue-Congo breakup    |
| Yoruba–Igbo      | 0.105 | 5,250                 | primary Volta-Niger split    |
| Yoruba–Mandinka  | 0.245 | 3,280                 | primary Niger-Congo branches |

The Yoruba–Kiswahili estimate of  $\sim 3,700$  years precedes the expected Proto-Benue-Congo breakup (4,000–5,000 years BP), as it must: the pair sits *inside* that family. The deeper Yoruba–Igbo figure of  $\sim 5,250$  years falls at the base of the Volta-Niger node, again as it must.

## 4.3 Indo-European

Table 4: Indo-European divergence estimates.

| Pair                       | $C$  | $\hat{t}$ (years ago) | Reference                           |
|----------------------------|------|-----------------------|-------------------------------------|
| Italian–French             | 0.67 | 1,850                 | Vulgar Latin breakup                |
| English–Swedish            | 0.61 | 2,290                 | Proto-Germanic disintegration       |
| Old Norse–Modern Icelandic | 0.92 | $\sim 380$            | extreme conservatism (floor effect) |

The Old Norse–Modern Icelandic row is the method’s control for near-identity. In spite of roughly a millennium of elapsed time the pair is nearly the same language, with  $C = 0.92$  far above any family-typical value. The estimator returns  $\sim 380$  years—shallow, and an *underestimate* of the elapsed millennium, precisely because Icelandic is

an extreme conservative outlier. The control shows the instrument’s floor: it refuses to push a close, thin divergence deep into the past, and any residual error runs in the conservative direction.

#### 4.4 Comparative summary

Table 5: Estimator versus historical reference across all validated pairs. All estimates agree with the independent historical envelope to within the stated uncertainty (§6); the sole value leaning outside (Tigrinya–Amharic) is an upper bound, produced by Amharic’s accelerating drift, as argued in the companion Ethio-Semitic study.

| Pair                | Family        | $\hat{t}$ (yr) | Historical (yr) |
|---------------------|---------------|----------------|-----------------|
| Gikuyu–Kinyarwanda  | Bantu         | 1,500          | 1,400–1,600     |
| Luganda–Kinyarwanda | Bantu         | 830            | 800–900         |
| Tigrinya–Amharic    | Ethio-Semitic | 2,110          | 1,500–2,000     |
| Amharic–Arabic      | Ethio-Semitic | 1,890          | 1,800–2,100     |
| Yoruba–Igbo         | Niger-Congo   | 5,250          | 5,000–5,500     |
| English–Swedish     | Indo-European | 2,290          | 1,900–2,300     |
| Egyptian–Coptic     | Afro-Asiatic  | 2,510          | 2,400–2,600     |

## 5 Relationship to existing methods

Table 6: The method in the landscape of historical linguistics.

| Feature                        | Comparative | Lexicostatistics | Bayesian | This method |
|--------------------------------|-------------|------------------|----------|-------------|
| Consonant-only comparison      | No          | No               | No       | <b>Yes</b>  |
| Quantifies sampling error      | No          | No               | Yes      | <b>Yes</b>  |
| Family-specific calibration    | No          | No               | Yes      | <b>Yes</b>  |
| Accounts for expansion bursts  | No          | No               | No       | <b>Yes</b>  |
| Automated and fully replicable | No          | Partial          | Yes      | <b>Yes</b>  |

The comparative method [Guthrie, 1971] reconstructs sound systems with great rigor but yields relative not absolute chronology, and requires expert judgment at every step. Classical lexicostatistics is transparent but anchored to the failed universal constant [Ringe, 2016, Bird, 2018]. Bayesian phylogenetics [Grollemund et al., 2015] is powerful but computationally opaque. The present method occupies the deliberate middle ground: the rigor of a stated model with the transparency of a screen-sized calculation that anyone can audit by hand.

## 6 Uncertainty

Because  $C$  is an empirical proportion of  $n$  independent gloss comparisons, its standard error is

$$\sigma_C = \sqrt{C(1-C)/n}.$$

Propagating through the estimator and neglecting the (smaller) parameter error,

$$\sigma_{\hat{t}} \approx b\tau \frac{\sigma_C}{|C \ln r|}.$$

For typical values ( $C = 0.6$ ,  $n = 100$ ,  $r = 0.75$ ) this is on the order of one to two centuries—an honest statement of precision, and one that should accompany every point estimate. We emphasize that the dominant sources of error in deep-time dating are *parameter* error and model misspecification, not sampling error; the reported agreement with historical anchors is therefore best read as evidence that parameter error is modest in the tested families.

## 7 Assumptions and limitations

The method rests on four assumptions that should be stated and checked openly:

1. *Rate stationarity within a family.*  $r_{\mathcal{F}}$  is common to all pairs in the family. Families whose expansion history is heterogeneous require sub-family calibration.
2. *Comparability of lists.* The  $n$  glosses must be translation-equivalent, which is the standing discipline of the Swadesh list [Swadesh, 1952].
3. *Documented correspondences.* Sound shifts are supplied rather than inferred; the results inherit any error in the shift system.
4. *A dateable anchor.* The calibration points (Table 1) themselves carry historical uncertainty, which is transferred directly into  $r$  and  $b$ .

## 8 Conclusion

The consonantal-skeleton method formalizes a simple observation: that the durable sediment of a language is its consonantal structure, and that a carefully calibrated comparison of that structure yields divergence times that agree with what history and archaeology already know. The estimator

$$\hat{t} = b\tau \frac{\ln C}{\ln r}$$

is derived from an explicitly stated retention model, fixed by closed-form calibration on dated anchors, and validated in five families spanning some of the deep and shallow chronology documented in African, European, and Near Eastern history. Because the pipeline is algorithmic, every number in this paper can be recomputed by any individual with a word list and a calculator; because the model is stated, the method is falsifiable. It offers, we believe, a transparent and rigorous instrument for questions—linguistic pre-history, oral-tradition history, and the timing of demographic collapse and expansion—where scholars have previously had to choose between rigor without transparency and transparency without rigor.

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